

Anthony Hernandez

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Education

University of California, Riverside

B.S. in Mechanical Engineering

Riverside, CA

Expected June 2028

- **Honors:** Dean's List (3x); NASA Community College Aerospace Scholar (NCAS); NSF AT3 Technician Fellow
- **Coursework:** Statics, Mechanics of Materials, Materials Science, Engineering Graphics and Design, Computational Modeling (MATLAB), Differential Equations, Linear Algebra, DC Circuits

Skills

CAD and Drafting: SolidWorks (CSWA certified), Fusion 360, GD&T, detailed part and assembly drawings

Simulation and Analysis: SolidWorks Simulation (FEA, CFD, topology optimization), ANSYS, RPA, NASA CEA

Programming and Data: MATLAB, Microsoft Excel (engineering calculators, sizing models)

Hands-On: Test stand setup, cold flow testing, 3D printing, engine and HVAC system service, wiring diagnostics

Experience

Co-Lead, Propulsion Team

Feb 2026 – Sept 2026

Norco Rocketry, Norco College

Norco, CA

- Co-led a 4-member team designing a liquid oxygen and ethanol rocket engine targeting 8,000 N thrust over a 10 s burn
- Designed a coaxial pintle injector and modeled the full engine in SolidWorks, authoring all GD&T drawings
- Built an Excel injector sizing model, verifying hand calculations and O/F ratio with RPA and NASA CEA
- Led material selection and ran stress and bolt load calculations that set fastener count at a safety factor of 2
- Presented injector, nozzle, and ablative liner designs at design reviews, balancing performance, cost, and machinability

Student Researcher, Propulsion Team

June 2025 – Aug 2025

Bronco Star Project (THRUST), Cal Poly Pomona

Pomona, CA

- Ran trade studies on cold gas, monopropellant, electric, and green propulsion for a NASA CubeSat proposal
- Applied NASA Systems Engineering Handbook methods to rank concepts on thrust, power, volume, and hazard class
- Evaluated propellant storage, feed systems, and safety constraints for rideshare launch and payload integration

Automotive Technician Assistant

Summers 2024 and 2025

JB Diesel Performance

San Bernardino, CA

- Completed 30 oil changes and 15 A/C services using refrigerant recovery, evacuation, leak-test, and recharge equipment
- Supported A/C system replacement and transmission, clutch, and flywheel removal under a supervising technician

Member, Propulsion Team

Aug 2024 – Apr 2025

Norco Rocketry, Norco College

Norco, CA

- Owned the injector for an N₂O and E85 engine, trading impinging, showerhead, and pintle designs on cost
- Retuned orifice size, count, and layout of a proven injector design, sizing it with a custom Excel calculator
- Built a water cold flow test stand, verifying spray pattern, atomization, and valve timing against hand calcs
- Supported 3 static-fire tests in the Mojave Desert, documenting valve, ablative, and control-system failures for redesign

Projects

5-Inch FPV Drone Frame | SolidWorks, FEA, CFD, Topology Optimization

July 2026 – Present

- Reduced frame weight 65% through topology optimization, validating the design with FEA and CFD in SolidWorks
- Refining the optimized frame for 3D printing ahead of physical flight testing

BMW E30 Engine Swap and Restoration | Personal Project

June 2024 – Dec 2025

- Swapped a 1985 E30 from the M20 eta to a higher-output M20, raising output 42% (120 to 170 hp)
- Integrated fuel, cooling, ignition, and A/C systems, diagnosing wiring faults, vacuum leaks, and fitment issues
- Researched part compatibility, tolerances, and conversion requirements using OEM technical documentation